SPECIFICATION — PRE-PAGE

NODE-SPECIFIC DISCLOSURE OVERVIEW (NON-LIMITING)

A. Purpose of Node-Specific Disclosure

This specification describes a modular multi-node system for privacy-preserving measurement, interpretation, and correlation of intimate physiology and health context. In addition to system-level descriptions, the disclosure includes node-specific sections intended to support standalone protectability of each node as an independent device and workflow, while also supporting optional multi-node correlation when used together.

B. Standalone Protectability Principle

Each node described herein is disclosed with sufficient detail to support embodiments where: 1) the node operates independently to acquire signals, compute validity/quality indices, and generate privacy-preserving outputs; and 2) the node optionally participates in cross-device or cross-user correlation under consent gating. Accordingly, any node may be implemented, manufactured, marketed, or deployed as an individual product without requiring implementation of other nodes.

C. General Structure of Each Node Section

Each node section is organized to include, in non-limiting form: 1) summary of purpose and technical field; 2) multiple physical embodiments and form factors; 3) sensor suites with substitution and equivalents; 4) data collection modes and sampling strategies; 5) validity gating, quality indices, and artifact rejection; 6) privacy-preserving output formats and example metric families; 7) anti-spoof and validation mechanisms; 8) standalone use cases and optional integration hooks; and 9) variations, substitutions, and combinations.

D. Non-Limiting Node List

The following nodes are described in dedicated sections, without limitation: Node 1: External longitudinal wearable configured for time-series physiology monitoring. Node 2: Intraluminal longitudinal wearable configured for time-series tissue-adjacent monitoring. Node 3: External session-based geometry scanner configured to generate baseline references. Node 4: Intraluminal geometry and/or compliance device configured for distance, contact, and stimulus-response characterization. Node 5: Upper-body context node configured to provide physiologic and activity context for interpretation and gating.

E. Incorporation of Appendices

Each node section may incorporate and apply the non-limiting principles described in the appendices, including sensor modality substitutions, sampling modes, quality indices, consent gating, privacy-preserving outputs, and validation/anti-spoof mechanisms.

F. Variations and Interpretation

Any node section may omit, add, or substitute sensor modalities and structural features while maintaining functional intent. Any described embodiment may be combined with any other embodiment unless expressly stated otherwise. The terms including and comprising are open-ended and do not exclude additional elements.